# Math for Economists Bootcamp

## Module 2, Part 1: Calculus and Optimization

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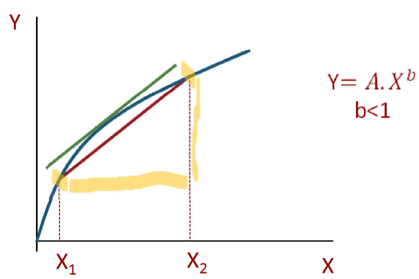
## Derivative

- **Derivative** – the slope of the tangent line to a function at a point, calculated by taking the limit of the difference quotient, is the derivative




- $y = f(x)$
- $\frac{\Delta y}{\Delta x} = \frac{f(X_2) - f(X_1)}{(X_2 - X_1)}$
- $\frac{\Delta y}{\Delta x} = \frac{f(X_1 + \Delta x) - f(X_1)}{\Delta x}$
- $\lim_{\Delta x \to 0} \frac{f(X_1 + \Delta x) - f(X_1)}{\Delta x}$
- Example: $y = mx + b$
  - $\lim_{\Delta x \to 0} \frac{m(X_1 + \Delta x) + b - mX_1 - b}{\Delta x}$
  - $\lim_{\Delta x \to 0} \frac{(m\Delta x)}{\Delta x} = m$

- **Differentiation** – the process of taking a derivative

## Calculus rules

- *If c is a constant:* $\frac{dc}{dx}=0$
- *If c is a constant: f: c*f(x)*
  $\rightarrow f'(x) = c * f'(x)$
- $f(x) = mx^n + c$
  $\rightarrow f'^{(x)} = m.n.x^{n-1}$
- $y(u)$, where $u = f(x)$
  $\rightarrow \frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$ (chain rule)
- $y = e^x \rightarrow \frac{dy}{dx} = e^x$
- $y = \ln(x) \rightarrow \frac{dy}{dx} = \frac{1}{x}$

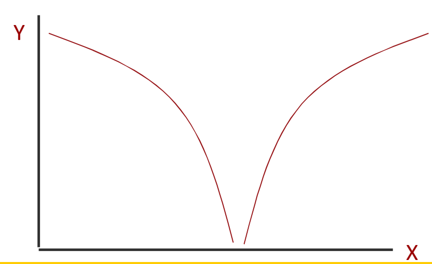
- $h(x) = g(x) + f(x)$
  $\rightarrow h'(x) = g'(x) + f'(x)$
- $h(x) = \sum_i g_i(x)$
  $\rightarrow h'(x) = \sum_i g_i{}'(x)$
- $h(x) = g(x).f(x)$
  $\rightarrow h'(x) = g'(x).f(x) + f'(x).g(x)$
- $h(x) = g(x)/f(x)$
  $\rightarrow h'(x) = \frac{g'(x).f(x) - f'(x).g(x)}{[f(x)]^2}$

**Chain Rule** - the chain rule defines the derivative of a composite function as the derivative of the outer function evaluated at the inner function times the derivative of the inner function



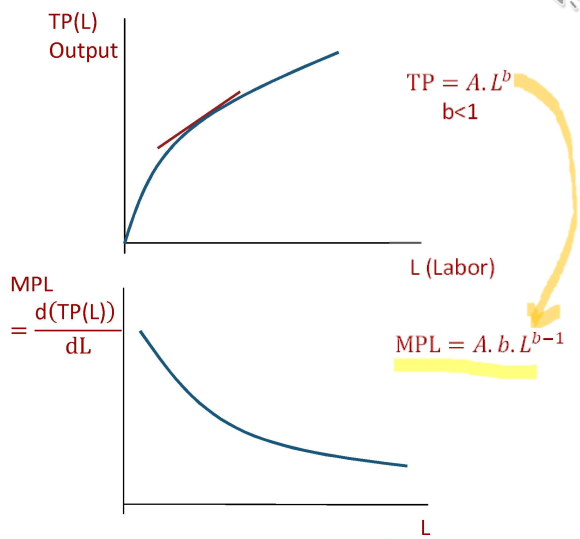
## Existence, Uniqueness and solutions

- In economics we usually use differentiable functions
- **Differentiable function** - a function for which f ′(x) exists
- If $f(x)$ is a continuous function then $x$ is a non-empty and closed set; and if bounded then there will be both a minimum and a maximum

# Derivatives and Economics

- Single variable
  - $f(x) = ax^b$
    - $f'(x) = \frac{df(x)}{dx} = a.b.x^{b-1}$

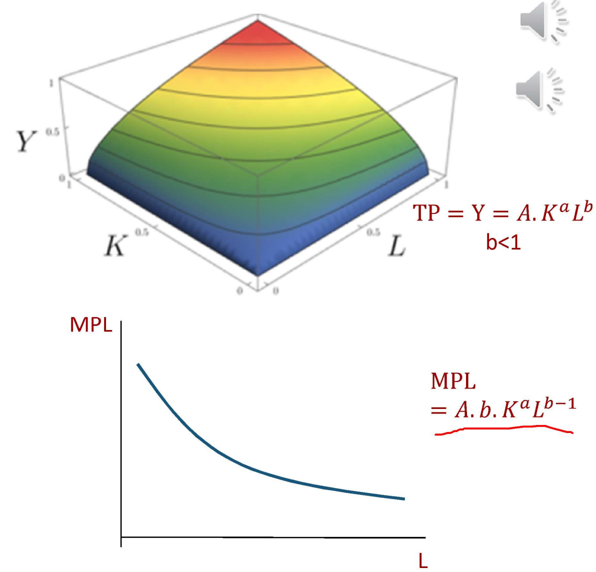





# Derivative

- Multiple variables
  - Partial derivative (marginal in economics)
  - $f(x,y) = x^a.y^b$
    - $f_x = \frac{\partial f(x,y)}{\partial x} = a.x^{a-1}.y^b$
    - $f_y = \frac{\partial f(x,y)}{\partial y} = b.x^a.y^{b-1}$
    - Store gradient of $f(x,y)$ into a vector

$$\nabla f = \begin{bmatrix} f_x \\ f_y \end{bmatrix} = \begin{bmatrix} f_L \\ f_K \end{bmatrix} = \begin{bmatrix} A.b.K^aL^{b-1} \\ A.a.K^{a-1}L^b \end{bmatrix}$$



## Examples

- $f(x) = 10x^3 + \cancel{4}$

$$\mathbb{R}_1 \to \mathbb{R}$$
$$f'(x) = 30x^2$$

- $f(x, y) = 5.x^4.y$

$$\mathbb{R}_2 \to \mathbb{R}$$
$$f_x = 20.x^3.y$$
$$f_y = 5.x^4.\underbrace{y^0}_{=1} = 5.x^4$$



## Examples

- $f(x, y) = (\overbrace{1 + 5.x^4}^{g(x)}).(\overbrace{y - 3}^{f(x)})$

$$f_x = (y - 3).20x^3 + (1 + 5.x^4).0 = (y - 3).20x^3$$
$$f_y = (y - 3).0 + (1 + 5.x^4).1 = (1 + 5.x^4)$$

Rule: $h(x) = g(x).f(x)$
$\to h'(x) = g'(x).f(x) + f'(x).g(x)$

## Implicit Differentiation

- a technique for computing dy/dx for a function defined by an equation, accomplished by differentiating both sides of the equation (remembering to treat the variable y as a function) and solving for dy/dx
- Example: assume $f(x,y) = x^2 + y^2 = 9$, find $\frac{dy}{dx}$ without rearranging for y first:

$$\frac{d(x^2+y^2)}{dx} = \frac{d(9)}{dx}$$

$$\frac{d(x^2)}{dx} + \frac{d(y^2)}{dx} = \frac{d(9)}{dx}$$

$$\frac{d(x^2)}{dx} + \frac{d(y^2)}{dy}\frac{dy}{dx} = \frac{d(9)}{dx}$$

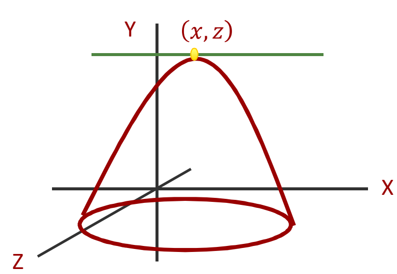
$$2x + 2y.\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-2x}{2y} = \frac{-x}{y}$$



## Maximum and Minimum

- At a maximum or a minimum (x,z)
  - $\nabla f = \begin{bmatrix} f_x \\ f_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
- Second order derivatives (Hessian)
  - $\nabla^2 F = \begin{bmatrix} f_{xx} & f_{xz} \\ f_{zx} & f_{zz} \end{bmatrix}$
  - Hessian matrix determines concavity/convexity



## Examples: second order

- $f(x, y) = 5.x^4.y$

$$f_x = 20.x^3.y$$
$$f_y = 5.x^4.y^{1-1} = 5.x^4$$

$$f_{xx} = 20.3.x^{3-1}.y = 60.x^2.y$$
$$f_{xy} = 20.x^3.y^{1-1} = 20.x^3$$
$$f_{yx} = 5.4.x^{4-1} = 20.x^3$$
$$f_{yy} = 0$$

$$\begin{tikzcd}[column sep=small] & & & f(x,y) \arrow[dl] \arrow[dr] & & & \\ & & f_x \arrow[dl] \arrow[dr] & & f_y \arrow[dl] \arrow[dr] & & \\ & f_{xx} & & f_{xy} = f_{yx} & & f_{yy} & \end{tikzcd}$$

Young's theorem



## Helpful Concepts

- **First Order Conditions** (F.O.C.) and **Second Order Conditions** (S.O.C.)
  - Another way of expressing the *first derivative test* and *second derivative test* you will have seen in calculus (economist love renaming things!)
  - The first order condition determines if the objective function has a local extremum (max or min)
  - The second order condition determines if this extremum is a local minimum or maximum
  - *Note*: The objective function must be continuous and twice differentiable for both conditions to hold

## Example

- Find max or min of:
  - $5.x^2 + y^2 = 0$

$$f_x = 10.x = 0$$
$$f_y = 2.y = 0$$

  - Solution: (x,y) = (0,0)
- Max or min? second derivatives

$$f_{xx} = 10$$
$$f_{xy} = 0$$
$$f_{yx} = 0$$
$$f_{yy} = 2$$



## Second order derivatives (Hessian)

- Positive definite – concave up (minimum)
  - If all the determinants of the minors are positive then positive definite
  - E.g., Hessian of form: $\begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix}$
  - $a_{11}$, $\begin{vmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{vmatrix}$ and $\begin{vmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{vmatrix} > 0$
- Negative definite – concave down (maximum)
  - If determinants of the minors switch signs
  - E.g., Hessian of form: $\begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix}$
  - $f_{11} < 0$
  - $\begin{vmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{vmatrix} > 0$
  - $\begin{vmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{vmatrix} < 0$

- Minors are related to eigenvalues
- If neither positive or negative could be a saddle point: eigenvalues go in different directions.

## Example

- Max or min?

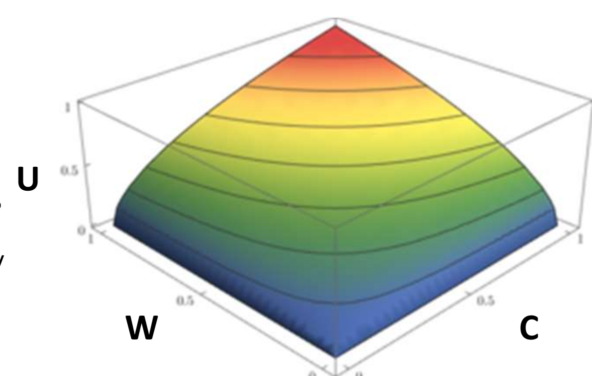
$$f_{xx} = 10$$
$$f_{xy} = 0$$
$$f_{yx} = 0$$
$$f_{yy} = 2$$

  – Hessian: $\begin{bmatrix} 10 & 0 \\ 0 & 2 \end{bmatrix}$
  – Positive definite - minimum



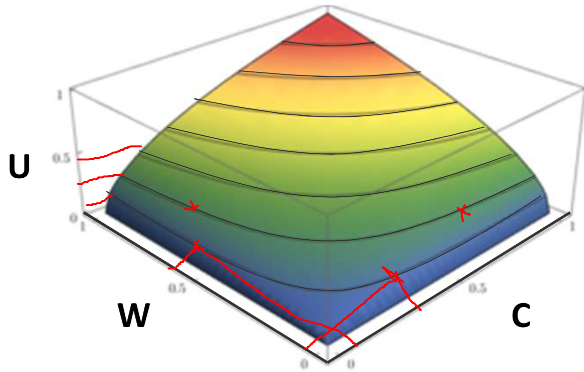
## Cobb Douglas

- Used for production and utility functions
  – $f(x_1, x_2) = {x_1}^{\alpha}.{x_2}^{\beta}$
  – production: $Q = f(K, L) = A.K^{\alpha}.L^{\beta}$
  – utility/happiness: $\mathrm{U} = f(W, C) = W^{\alpha}.C^{\beta}$
- Why do we like Cobb Douglas functions?
  – $f(x_1, x_2)$ rises with more $x_1$ $or$ $x_2$ (e.g., utility rises with more consumption of wheat and cloth)
  – But at a diminishing rate – the slope or first derivative is falling (i.e., diminishing marginal utility: your 50th bushel of wheat is less valuable than your 1st bushel)



## Level sets

- Want to draw in two dimensions (W and C)
- Level sets (horizontal slices)
  - Combinations of W and C at same height (i.e., constant utility)



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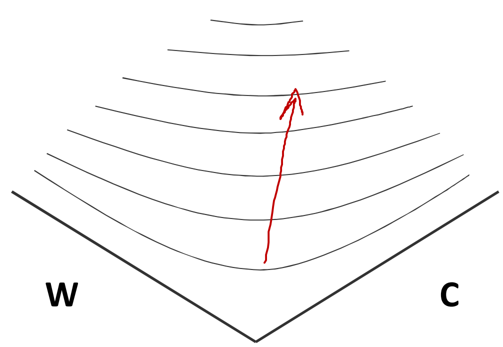
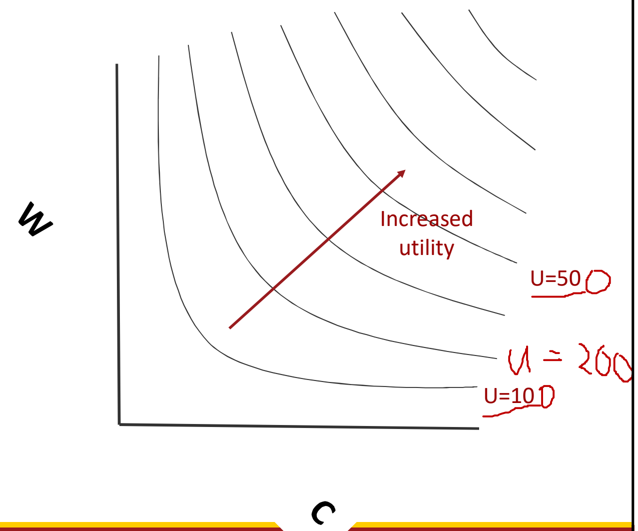


## Level sets

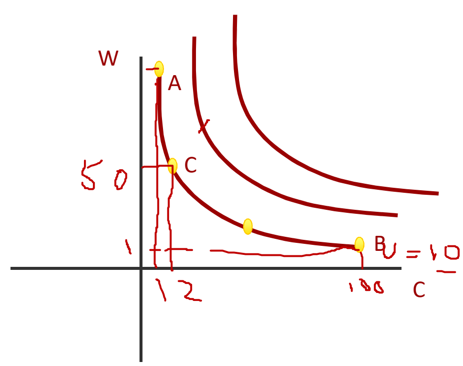
- Two dimensions (W and C)
- Level sets (horizontal slices)
  - Combinations of W and C at same height give constant utility (e.g., $\mathrm{U} = f(W, C) = 10$)





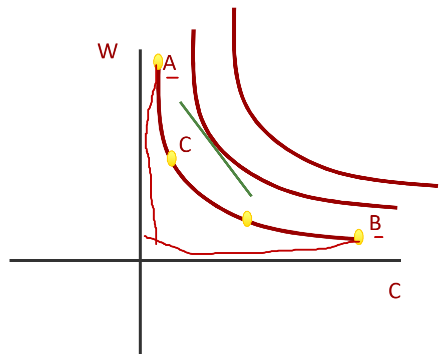

## Level sets

- Combinations of W and C that give you a given amount (e.g., $f(W, C) = 10$)
  - $A = 1, \alpha = \beta = \frac{1}{2}$
  - $f(W, C) = 1.W^{1/2}.C^{1/2} = 10$
  - $W = 1, C = 10^2 = 100$ (B)
  - $C = 1, W = 10^2 = 100$ (A)
  - $\mathrm{W} = 50, C = \left(\frac{10}{\sqrt{50}}\right)^2 = 2$ (C)



## Level sets

- Notice that along a level set (U= $f(W,C)$)
  - $\Delta(f(W,C)) = \Delta U = f_W.\Delta W + f_C.\Delta C = 0$
- Utility functions: indifference curves
  - Slope is MRS
  - $MRS = \frac{\Delta W}{\Delta C} = -\frac{MU_C}{MU_W} = -\frac{f_C}{f_W} = -\frac{\frac{dU}{dC}}{\frac{dU}{dW}}$
- Production functions: isoquants
  - Slope is the MRTS
  - $MRTS = \frac{\Delta K}{\Delta L} = -\frac{f_L}{f_K} = -\frac{\frac{dQ}{dL}}{\frac{dQ}{dK}} = -\frac{MPL}{MPK}$






## Appendix: Helpful Definitions

- **Chain Rule** - the chain rule defines the derivative of a composite function as the derivative of the outer function evaluated at the inner function times the derivative of the inner function
- **Derivative** – the slope of the tangent line to a function at a point, calculated by taking the limit of the difference quotient, is the derivative
- **Differentiation** – the process of taking a derivative
- **Differentiable function** - a function for which f '(x) exists
- **Higher-order derivative** – a derivative of a derivative, from the second derivative to the *n*th derivative, is called a higher-order derivative
- **Implicit Differentiation** – is a technique for computing dy/dx for a function defined by an equation, accomplished by differentiating both sides of the equation (remembering to treat the variable y as a function) and solving for dy/dx
- **Logarithmic Differentiation** - technique that allows us to differentiate a function by first taking the natural logarithm of both sides of an equation, applying properties of logarithms to simplify the equation, and differentiating implicitly